

Algebra — revision of Test 1

Lessons 1–8 · Quarter I · 60 questions

Work down each column. Where a fraction is cancelled, the permissible values are part of the answer — an answer without them is half an answer.

NAME	CLASS	DATE
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Lessons 1–3 · Revision of the Grade 7 course22

Indices, brackets, factorising

1 $x^7 \cdot x^5$	8Expand $3(2x - 7)$	16Expand $(3m - 2)^2$
2 $a^9 \div a^4$	9Expand $-5(a - 4)$	17Expand $(x - 6)(x + 6)$
3 $(b^4)^3$	10Expand $x(x + 6)$	18Factorise $8x + 20$
4 $(2x^3)^4$	11Expand $2a(3a - 5)$	19Factorise $6a^2 - 9a$
5 $(-3a^2b)^2$	12Expand $(x + 3)(x + 8)$	20Factorise $x^2 - 49$
6 $5x^2 \cdot 4x^5$	13Expand $(y - 5)(y + 2)$	21Factorise $25m^2 - 4n^2$
7 $\frac{12a^6}{3a^2}$	14Expand $(2x - 1)(x + 4)$	22Factorise $x^2 + 10x + 25$
	15Expand $(a + 7)^2$	

Lessons 4–5 · Algebraic expressions16

Powers of products, factorising, simplifying

23 $(2a^2b^3)^3$	27Factorise $3x^2 - 12$	31Factorise $ax + ay + bx + by$
24 $(-x^4y)^3$	28Factorise $2a^2 - 18b^2$	32Factorise $m^2 - m - 12$
25 $(5p^3)^2 \cdot 2p$	29Factorise $x^3 - x$	33Factorise $x^2 + 7x + 12$
26 $\frac{a^5b^2}{a^2b}$	30Factorise $4y^2 - 20y + 25$	34Factorise $5x^2y - 10xy^2$

- 35

Simplify $3(x - 2) - 2(x - 5)$
- 37

Simplify $(x + 2)(x - 2) + 4$

- 36

Simplify $(a + 4)^2 - (a - 4)^2$
- 38

Find $x^2 - 4x$ when $x = -2$

Lessons 6-8 · Algebraic fractions. Cancelling

22

Permissible values, value at a point, cancelling

- 39

Permissible values of $\frac{1}{x - 9}$
- 47

For which x is $\frac{x - 8}{x + 2}$ zero?
- 55

Cancel $\frac{x^2 + 6x + 9}{x^2 - 9}$
- 40

Permissible values of $\frac{5}{2x + 6}$
- 48

For which x is $\frac{3x + 6}{x - 1}$ zero?
- 56

Cancel $\frac{3a - 3b}{b - a}$
- 41

Permissible values of $\frac{x + 1}{x^2 - 4}$
- 49

Cancel $\frac{24x^5}{6x^2}$
- 57

Cancel $\frac{x^2 - x}{1 - x}$
- 42

Permissible values of $\frac{3}{x^2 + 1}$
- 50

Cancel $\frac{14a^3b}{7ab}$
- 58

Cancel $\frac{4x^2 - 9}{2x + 3}$
- 43

Permissible values of $\frac{x - 2}{x(x - 7)}$
- 51

Cancel $\frac{5x + 15}{x + 3}$
- 59

Cancel $\frac{x^3 - 4x}{x^2 + 2x}$
- 44

Find $\frac{x + 7}{x - 3}$ when $x = 5$
- 52

Cancel $\frac{x^2 - 16}{x - 4}$
- 60

Why can $\frac{x + 3}{3}$ not be cancelled?
- 45

Find $\frac{2x - 1}{x + 4}$ when $x = -1$
- 53

Cancel $\frac{x^2 - 25}{x^2 + 5x}$
- 46

Find $\frac{x^2 - 9}{x + 3}$ when $x = 4$
- 54

Cancel $\frac{2x + 8}{x^2 - 16}$